MoS2 shell thickness in TENGs

"How does MoS2 shell thickness affect TENG output?" *Generated by LeapSpace*

A thinner, continuous MoS_2 shell is the design supported by the retrieved TENG evidence: it preserves charge density, suppresses charge decay, and improves humidity resistance, while the evidence does not provide a direct thickness sweep showing an optimum for MoS_2 shell thickness in TENG output. [11]

What the evidence supports

Shell-thickness implication	Effect on TENG output	Evidence
Thin, uniform MoS_2 shell	Better output stability because the shell shields the SiO_2 interface from water and slows charge decay	The MoS_2-encapsulated SiO_2 TENG used a **thin, uniformly grown MoS_2 layer** to stabilize charge density and retain **more than 70% output over 25 h at 99% RH** [11]
Excessively thick shell	Not directly measured for this TENG system	No retrieved TENG source gives a thickness series for MoS_2 shell thickness on SiO_2 and output change [11]
Too little shell coverage	Likely weaker humidity protection and less charge confinement, by inference	This is an inference from the reported role of the shell as a hydrophobic, charge-stabilizing barrier; it is not directly quantified in the retrieved data [11]

Practical interpretation

The clearest takeaway is that **MoS_2 shell thickness should be just enough to form a continuous, conformal coating**. That geometry gives the shell its main TENG benefits: **humidity shielding, slower charge decay, and stable triboelectric output** [11].

The retrieved material does **not** show a direct “thicker is better” or “thinner is better” curve for MoS_2 shell thickness in TENGs, so any stronger claim would be an inference rather than a measured result [11].

Bottom line: for MoS_2-shell/SiO_2-core TENGs, the evidence supports **thin and uniform shell coverage** as the safest route to high and stable output; shell thickness beyond that is not quantified in the retrieved studies, so the exact optimum remains unsupported here [11].

Confidence level: **Medium** — the functional role of a thin MoS_2 shell is directly supported, but a true thickness–output optimization study for this exact TENG architecture was not retrieved.

References

11. Kim, D.-H., Park, J.Y., Choi, H.S., (...), Baik, J.M. (2025) Advancing Humidity-Resistant Triboelectric Nanogenerators Through MoS_2-Encapsulated SiO_2 Nanoparticles for Self-Powered Gas Sensing Applications. *Advanced Energy Materials*. https://doi.org/10.1002/aenm.202405278